FEDERAL SPECIFICATION

SEPARATOR, TEETH, DENTAL, ELLIOTT, CURVED-JAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a separator, used in the oral cavity to separate teeth during dental operative procedures.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–I–526 — Instruments, Dental and Surgical General Specifications.

Federal Standards:

Fed. Std. No. 66 — Steel; Chemical Composition and Hardenability.
Fed. Std. No. 123 — Marking for Domestic Shipments (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–P–12902 — Packaging of Surgical, Dental, and Veterinary Instruments, Diagnostic Equipment, and Related Items.

Miiltary Standard:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The separator components shall be fabricated of corrosion-resisting steel of the composition specified in table I below and shall be in accordance with the requirements of Federal Standard No. 66 for the composition specified. Material shall meet the requirements for chemical analysis when tested as specified in 4.4.1.

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main control. cycle-phase lights. chamber-vacuum-pressure gauge, sterilizing-

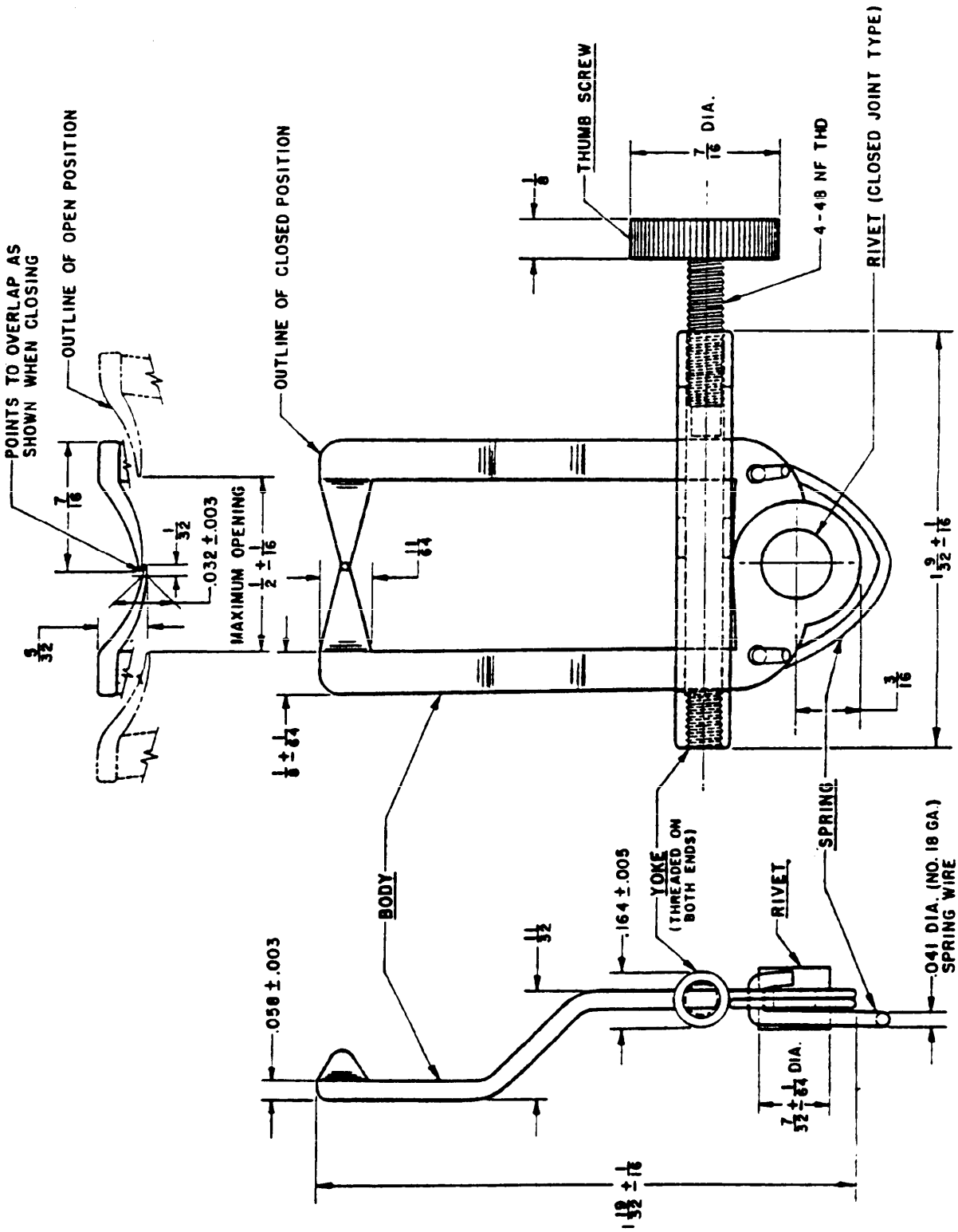


FIGURE 1. THUMB SCREW OMITTED FOR CLARITY

UNLESS OTHERWISE SPECIFIED TOLERANCE = ± 1/32

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TABLE I.

Component	Composition
Body	AISI 420
Rivet	AISI 303
Thumbscrew	AISI 416
Yoke	AISI 431
Spring	AISI 410

3.2 **Hardness.**—The jaws of the separator shall be hardened, tempered, electrolytically polished, and stress-relieved to a Rockwell hardness of C48–C53, when tested as specified in 4.4.2. The spring shall be hardened to the proper temper for its intended purpose.

3.3 **Style, design, and dimensions.**—Separator shall be the type known as the Elliott curved-jaw separator, posterior type. The wedge-shaped jaws shall be curved and tapered, so as to gain accessibility between interproximal surfaces of teeth and to prevent slipping or unnecessary trauma to the tissues as the thumbscrew is tightened. The thumbscrew shall operate in either side of the threaded yoke to provide for interchangeability of position in the dental arch. The spring shall be of sufficient tension and length for the separator to operate properly. Style, design, and dimensions of the separator shall be as shown in figure 1.

3.4 **Finish.** — The separator components shall be ground to remove all surface imperfections. The body, yoke, and spring shall be electrolytically polished, resulting in a mirror finish. Rivet and thumbscrew shall be polished to remove grind marks resulting in a mirror or satin finish. The finished instruments shall pass the copper sulfate and boil tests specified in 4.4.3.1 and 4.4.3.2 respectively.

3.5 **Identification marking.**—Each separator shall be marked with the name or registered trade-mark of the manufacturer and with the letters "E. SEP". All markings shall be legible and permanent.

3.6 **Workmanship.** — Separators shall be symmetrically made, smoothly and properly finished; moving parts shall work freely without binding. Separators shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.** —The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* — When available, certificates of quality, supplied by the manufacturer of the raw material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**

4.2.1 *For examination.* — Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

4.2.2 *For tests.* — Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, except that the ac-

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TABLE II.

Characteristic	Req.	Test procedure	Rqmt. appl.		Unit of product	Insp. level	Minimum sample size
			Indiv. unit	Lot avg.			
Hardness	3.2	4.4.2	X		1-separator	L2	2
Copper sulfate test	3.4	4.4.3.1	X		1-separator	L3	3
Boil test	3.4	4.4.3.2	X		1-separator	L3	3

ceptance number shall be zero for all sample sizes. (See table II.)

4.2.3 *Raw materials.* — Four ounces of each lot of each metal used in the construction of the separators shall be selected for chemical analysis.

4.2.4 Where possible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. — The separators shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. — Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Chemical analysis.* — In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of Federal Test Method Standard No. 151.

4.4.2 *Hardness test.* — Hardness shall be determined in accordance with method 243 (Rockwell) or method 244 (Vickers) of Federal Test Method Standard 151.

4.4.3 *Corrosion resistance tests.*

4.4.3.1 *Copper sulfate test.* — The separators shall be tested for copper sulfate corrosion resistance as specified in Federal Specification GG–I–526.

4.4.3.2 *Boil test.* — The separators shall be tested for boil test resistance as specified in Federal Specification GG–I–526.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 *Unit of issue.* — One separator, as specified, constitutes one unit.

5.1.2 *Quantities.* — The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE III. — *Packaging and packing quantities*

Unit package	*Intermediate package*	*Exterior container*
1 unit	12 units	288 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 *Level A.* — Packaging and packing shall conform to Military Specification MIL–P–12902, Procedure Code No. 35.

5.1.4 *Level B.* — Packaging and packing shall conform to Military Specification MIL–P–12902, Procedure Code No. 21.

5.1.5 *Level C.* — The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 **Marking—nonmilitary.**—In addition to any special marking specified in the contract or order shipping containers shall be in accordance with Federal Standard No. 123.

5.3 **Marking—Military.**—Marking shall be as specified in 3.6 through 3.6.2 of Military Specification MIL–P–12902, except that the ASMPA stock number shall not be required.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents shall specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 **Transportation description.** — Transportation description applicable to this item is:

Rail:

Dental Instruments

Carload minimum weight 12,000 pounds subject to Rule 34, Uniform Freight Classification.

Motor:

Dental Instruments, NOI

Truckload minimum weight (W) 10.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:

Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.

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